Long-Range Communication mesh for Disaster Response and Relief Coordination

ETAC 3252 Group Design Project

Group No: 06

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1. Introduction

i. Identified Problem

When big disasters like floods or landslides happen in world, the first things that usually break are the phone towers and power lines. This creates a "communication blackout" where people trapped in those areas can't call for help or tell anyone where they are. Right now, we depend too much on the internet and mobile signals. When those systems fail, there is no easy way for rescue teams to find victims

ii. Background Information

- **Advantages of Disaster Response Communication Systems**

The primary goal of a disaster response communication system is to provide a "lifeline" when traditional networks fail. An effective system ensures that real-time information can flow between victims and rescue teams, regardless of the state of the local power grid or physical infrastructure. The core advantages include high reliability, low power consumption [1] for long-term operation, and the ability to cover remote or geographically challenging terrain where signal blockages are common. By enabling rapid coordination, these systems significantly reduce the time taken to locate survivors and distribute emergency aid.

- **Currently Available Systems**

In Sri Lanka, the current disaster communication landscape relies heavily on the Disaster Management Centre (DMC) and its established protocols. These include:

- **Public Cellular Networks (4G/5G):** Used for mass SMS alerts (DEWN system) and social media coordination.
- **Satellite Technology:** Employed by high-level government agencies for specialized emergency links.
- **VHF/UHF Radio:** Utilized primarily by the military, police, and amateur radio societies (RSSL) for point-to-point voice communication.
- **Early Warning Towers:** A network of physical towers designed to sound sirens and broadcast announcements in coastal and high-risk areas.

- **Things to Improve in Currently Available Systems**

While the existing systems are robust under normal conditions, several critical gaps remain that need to be addressed:

- **Infrastructure Independence:** Infrastructure Independence: Most current systems are highly dependent on cellular towers and national electricity grids. During major catastrophic events, power outages and physical damage to fiber-optic backbones can render these systems useless, creating total "dead zones.".

- **Network Congestion:** During a crisis, mobile networks often become "saturated" by high traffic volumes, preventing emergency messages from reaching their destination.
- **Range and Interoperability:** Short-range technologies like WiFi are insufficient for large-scale disasters, while professional radio equipment is often too expensive or complex for the average citizen to use.
- **Data-Driven Coordination:** There is a need for a decentralized "mesh" approach where messages can "hop" from one device to another. This would eliminate "dead zones" caused by mountains or destroyed towers, ensuring that a single point of failure does not bring down the entire communication network.

iii. Aim of the Project

This project aims to build an independent disaster response communication system for disaster zones where all mobile and internet services have failed. It provides a reliable way for victims to send SOS alerts and GPS locations [2] while also enabling rescue teams to communicate and coordinate their efforts in the field.

iv. Objectives

- To compare available communication technologies such as LoRa, Wi-Fi, Bluetooth, and cellular networks, and identify the most suitable technology for disaster communication.

- To design and develop a portable communication device using the selected technology that can operate during emergency situations.

- To implement an efficient mesh-based communication system that allows devices to transmit, receive, and relay messages between nodes.

2. Literature Review

- Traditional Cellular and Satellite Networks
  The most common way people communicate today is through 4G/5G mobile networks or satellite phones. While these systems are very fast, they depend on expensive and complex infrastructure like towers and satellites. During a major disaster, cellular towers often lose power or get physically damaged, leading to a "dead zone". Satellite phones are a possible backup, but they are very expensive to buy and maintain, making them unavailable for the average person or local rescue teams in Sri Lanka.

- Short-Range Wireless Technologies (WiFi & Bluetooth)
  WiFi and Bluetooth are standard in all modern smartphones, making them the easiest way for victims to connect to a network. However, their range is usually under 100 meters, which is too short to cover a large disaster area. Using only WiFi for long distances would require too many repeaters and too much power, which is not possible during a total blackout.

- Long Range (LoRa) Communication

  LoRa (Long Range) is a low-power wireless technology that has recently become a popular choice for emergency systems. [3] It can send small data packets over 5-10 kilometers in open areas without needing any infrastructure. Unlike WiFi, LoRa uses much less energy, allowing nodes to run for days on small batteries. However, a single LoRa link is "point-to-point," meaning it only works if the two devices are within a certain distance. If a mountain or a building blocks the signal, the connection is lost.

Technology	Range	Infrastructure Dependence	Power Consumption	Cost & Accessibility
4G / 5G Cellular	High(Global)	Very High (Towers)	Moderate	Subscription-based; High
Satellite	Very High	High (Satellites)	High	Very Expensive
Wi-Fi / Bluetooth	Low(<100m)	Low(Local Routers)	Moderate	Built-in(Free)
LoRa	High(5-10Km)	None(Independent)	Very Low	Low Cost

Table 1

3. Methodology

Firstly, a literature review will be conducted to identify existing disaster communication technologies. The advantages and disadvantages of these technologies will also be identified. The technologies to be reviewed include LoRa, Wi-Fi, Bluetooth, and cellular networks. The suitability and limitations of these technologies will be identified, and the most appropriate technology will be chosen for this project.

Secondly, the portable communication device will be designed using the chosen technology. The technology chosen will be used to design the portable communication device. The system architecture will be designed, and the hardware components will be chosen. The device will then be designed in such a way that it is compact and portable.

Thirdly, the prototype of the device will be designed. The hardware components will be assembled and programmed using the chosen technology. The microcontroller will then be programmed in order to enable message transmission and reception.

Lastly, a mesh-based communication system will be designed using the developed device. The device will then be designed in such a way that it can transmit and receive messages to and from other devices. The device will then be used to create a mesh-based communication system, whereby the device will be able to transmit and receive messages to and from other devices, thus enhancing communication coverage.

4. Block Diagram

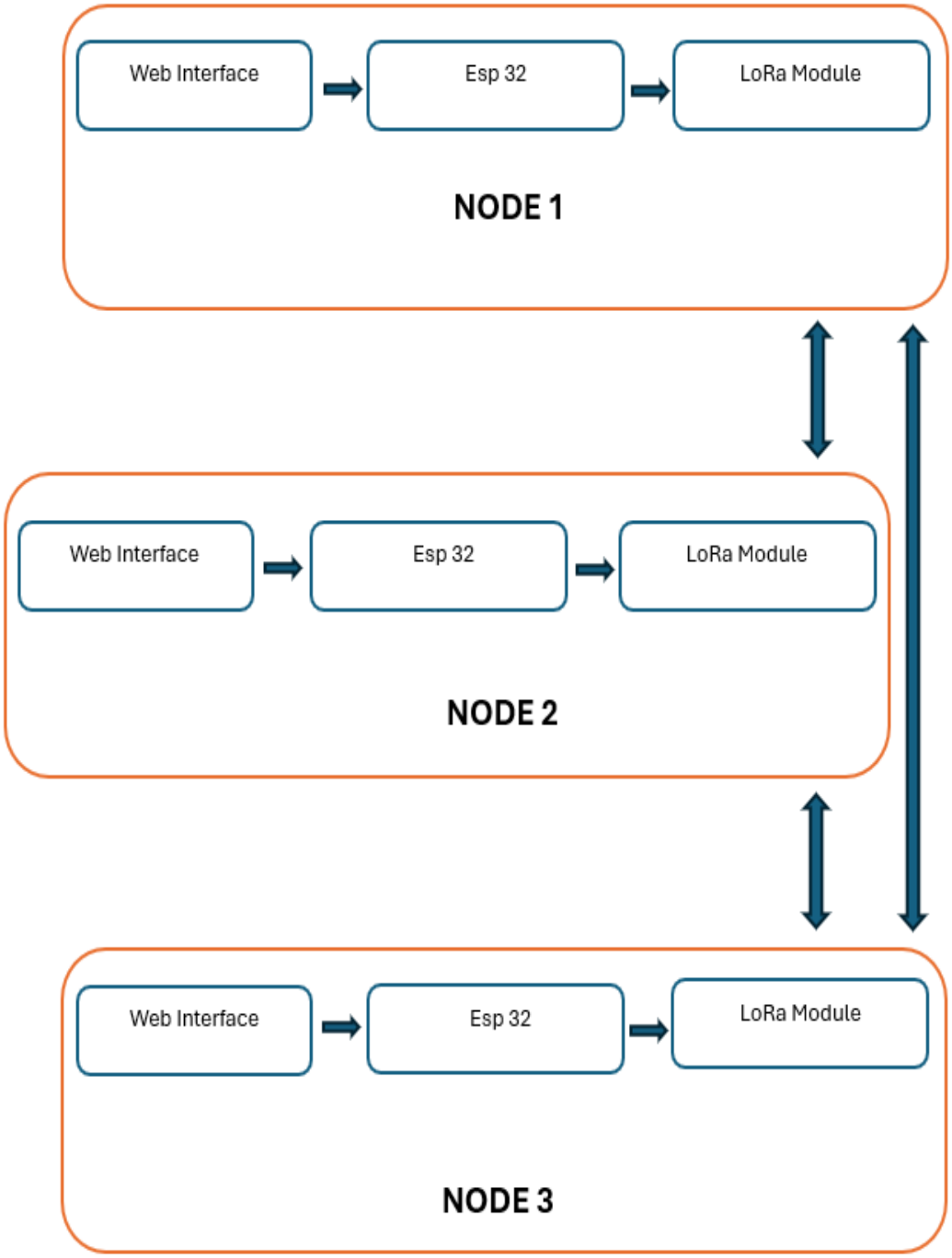

Figure 1

5. Budget Estimate

The estimated budget for the Long-Range Communication mesh for Disaster Response and Relief Coordination System project is as follows:

	Item	Quantity	Unit price	Total
01)	SX1278 LoRa Module	3	1490	4470
02)	ESP 32 Development Board	2	1290	2580
03)	1800mAh 3.7V 18650 Battery	4	460	1840
04)	TP4056 Charging Module	3	115	345
05)	AMS1117 5V to 3.3V Step down Regulator	3	70	210
06)	Active Buzzer 5V	2	50	100
07)	2N2222 NPN 40V 0.8A Transistor	5	6	30
08)	220 Ohm Resistor 10PCS	2	15	30
09)	1K Ohm Resistor 10PCS	1	15	15
10)	2Pin Rocker Switches	3	40	120
11)	Dot Board	1	275	275
12)	18650 Battery Holder	3	110	330
13)	Female Pin Header 2.54mm 1x40	10	30	300
14)	10uF 25V Aluminum Electrolytic Capacitor	8	5	40
15)	100uF 35V Aluminum Electrolytic Capacitor	5	10	50
16)	1N5818 Schottky Diodes	3	7	21
17)	Jumper Wires 40PCS	2	190	380
18)	Box Enclosure Case	3	690	2070
19)	433MHz SMA Folding Antenna	3	400	1200
				14406

Table 2

6. Activity Plan

Task	February		March				April				May			
	3rd week	4th week	1st week	2nd week	3rd week	4th week	1st week	2nd week	3rd week	4th week	1st week	2nd week	3rd week	4th week
Chose a problem to solve														
Literature review and communication method comparison														
Create project proposal														
Finalize system design														
Hardware component purchasing														
Device assembly and circuit design														
Programming microcontroller and LoRa communication														
Laboratory testing														
Field testing														
Performance analysis														
Report writing														
Final presentation preparation														

Table 3

7. References

[1] "ESP32 Series Datasheet," 2024. [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf. [Accessed 2026 March 26].

[2] "NEO-6 u-blox 6 GPS Modules Data Sheet," [Online]. [Accessed 08 March 2026].

[3] "LoRa Module 433 MHZ 10KM Ra-02," 2026. [Online]. Available: https://mikroelectron.com/product/lora-module-433mhz-10km-ra-02-ai-thinker-wireless-transmission?srsltid=AfmBOorSwdWw0d1auGeFzeH0x5VkqzwLtV7ZzcfhC_s07kVF2HCF32Oh.